

LAPORAN PRAKTIKUM 15 SEMENTARA

SCIKIT-LEARN

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1. Pendahuluan :

Perkembangan teknologi machine learning memungkinkan komputer untuk melakukan proses pengenalan pola secara otomatis berdasarkan data yang telah dipelajari sebelumnya. Salah satu library yang banyak digunakan dalam implementasi machine learning adalah **Scikit-learn**, yang menyediakan berbagai algoritma klasifikasi, regresi, serta evaluasi model secara mudah dan efisien menggunakan bahasa pemrograman Python.

Pada praktikum ini dilakukan proses klasifikasi pose tubuh manusia dengan memanfaatkan algoritma machine learning yang tersedia pada Scikit-learn, yaitu Random Forest Classifier. Sistem dirancang untuk mengenali dua jenis pose tubuh, yaitu berdiri dan duduk. Sebelum proses pelatihan model dilakukan, data pose tubuh dikumpulkan menggunakan webcam dan kemudian diproses untuk mendapatkan fitur berupa koordinat titik-titik tubuh.

Fitur tersebut digunakan sebagai dataset yang selanjutnya dilatih menggunakan algoritma klasifikasi pada Scikit-learn. Setelah model selesai dilatih, sistem diuji untuk mengetahui tingkat akurasi klasifikasi yang dihasilkan. Model yang telah memiliki performa baik kemudian diimplementasikan pada sistem deteksi pose secara real-time sehingga komputer dapat memprediksi pose tubuh manusia secara otomatis.

Dengan adanya proses ini, praktikum tidak hanya berfokus pada pengambilan data citra, tetapi lebih menekankan pada bagaimana machine learning khususnya Scikit-learn digunakan untuk membangun sistem klasifikasi berbasis data.

2. Tujuan Praktikum :

- a. Mempelajari konsep dasar machine learning dalam proses klasifikasi data.
- b. Menggunakan library Scikit-learn untuk membangun model klasifikasi pose tubuh.
- c. Mengolah dataset yang diperoleh dari hasil ekstraksi fitur tubuh.
- d. Melatih dan menguji model machine learning menggunakan algoritma Random Forest.
- e. Mengimplementasikan model klasifikasi ke dalam sistem deteksi pose secara real-time

3. Manfaat Praktikum

- a. Mahasiswa memahami penerapan machine learning menggunakan Scikit-learn.
- b. Meningkatkan kemampuan dalam proses pelatihan dan pengujian model klasifikasi.
- c. Memahami alur pembuatan sistem pengenalan pola berbasis dataset.
- d. Memberikan pengalaman dalam implementasi model machine learning pada aplikasi nyata.
- e. Menjadi dasar dalam pengembangan sistem klasifikasi berbasis computer vision dan machine learning

4. Hasil Program

A. Pengumpulan data menggunakan webcam

```
1  import os
2  from time import sleep
3  import cv2
4
5  DATA_DIR = r'C:\Users\user\Documents\Kontrol Cerdas\DATA'
6
7  if not os.path.exists(DATA_DIR):
8      os.makedirs(DATA_DIR)
9
10 number_of_classes = 2
11 dataset_size = 100
12
13 cap = cv2.VideoCapture(0)
14
15 for j in range(number_of_classes):
16     if not os.path.exists(os.path.join(DATA_DIR, str(j))):
17         os.makedirs(os.path.join(DATA_DIR, str(j)))
18     print('Collecting data for class {}'.format(j))
19     while True:
20         ret, frame = cap.read()
21         cv2.putText(frame, text='Ready? Press "Q" !', org=(100, 50),
22                     cv2.FONT_HERSHEY_SIMPLEX, fontScale=1, color=(0, 255, 0), thickness=3)
23         cv2.imshow(winname='frame', frame)
24         if cv2.waitKey(15) & 0xFF == ord('q'):
25             break
26     counter = 0
27     while counter < dataset_size:
28         ret, frame = cap.read()
29         cv2.imshow('frame', frame)
30         cv2.waitKey(10)
31
32         file_path = os.path.join(DATA_DIR, str(j), f'{counter}.jpg')
33
34         cv2.imshow('frame', frame)
35         cv2.waitKey(10)
36
37         file_path = os.path.join(DATA_DIR, str(j), f'{counter}.jpg')
38         cv2.imwrite(file_path, frame)
39
40         counter += 1
41 cap.release()
42 cv2.destroyAllWindows()
```

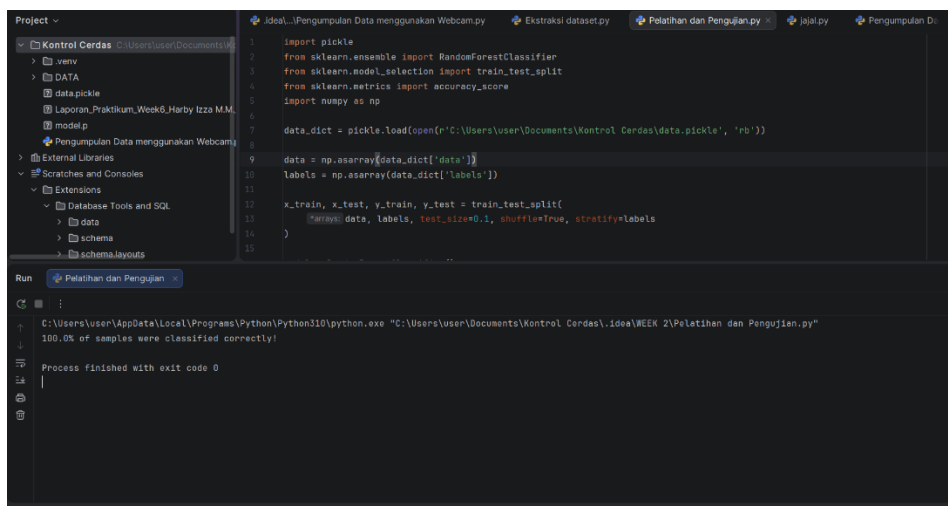
B. Ekstraksi dataset

```
1 import os
2 import pickle
3 import mediapipe as mp
4 import cv2
5
6 mp_hands = mp.solutions.hands
7 hands = mp_hands.Hands(static_image_mode=True, max_num_hands=1)
8 DATA_DIR = 'C:\\Users\\Documents\\Kontrol Cerdas'
9 data = []
10 labels = []
11
12 for dir_ in os.listdir(DATA_DIR):
13     for img_path in os.listdir(os.path.join(DATA_DIR, dir_)):
14
15         data_aux = []
16         x_ = []
17         y_ = []
18
19         img = cv2.imread(os.path.join(DATA_DIR, dir_, img_path))
20         img_rgb = cv2.cvtColor(img, cv2.COLOR_BGR2RGB)
21
22         results = hands.process(img_rgb)
23
24         if results.multi_hand_landmarks:
25             for hand_landmarks in results.multi_hand_landmarks:
26
27                 for i in range(len(hand_landmarks.landmark)):
28                     x = hand_landmarks.landmark[i].x
29                     y = hand_landmarks.landmark[i].y
30                     x_.append(x)
31                     y_.append(y)
32                 for i in range(len(hand_landmarks.landmark)):
33                     x_.append(x)
34                     y_.append(y)
35                 for i in range(len(hand_landmarks.landmark)):
36                     data_aux.append(x_[i] - min(x_))
37                     data_aux.append(y_[i] - min(y_))
38
39                 data.append(data_aux)
40                 labels.append(dir_)
41
42 f = open('data.pickle', 'wb')
43 pickle.dump({'data': data, 'labels': labels}, f)
44 f.close()
```

Sehingga dihasilkan file tersebut

 data.pickle	2/25/2026 9:03 PM	PICKLE File	76 KB
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C. Pelatihan dan Pengujian

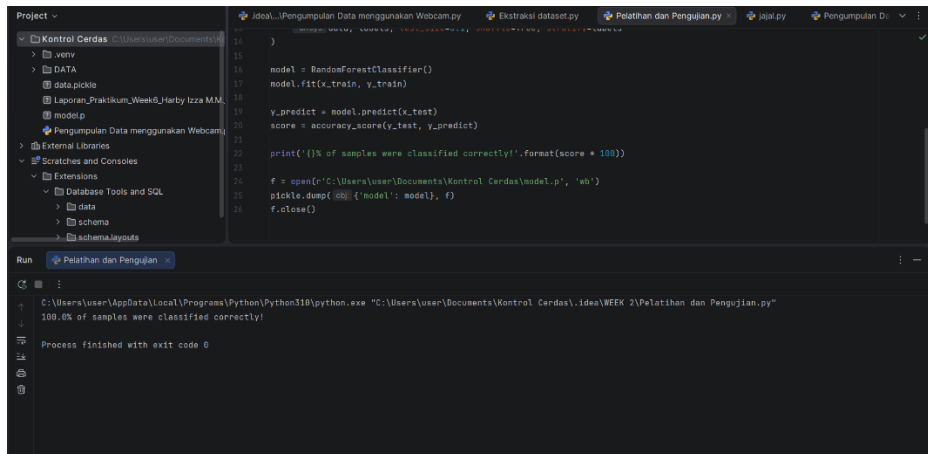


```
1 import pickle
2 from sklearn.ensemble import RandomForestClassifier
3 from sklearn.model_selection import train_test_split
4 from sklearn.metrics import accuracy_score
5 import numpy as np
6
7 data_dict = pickle.load(open('C:\\Users\\Documents\\Kontrol Cerdas\\data.pickle', 'rb'))
8 data = np.asarray(data_dict['data'])
9 labels = np.asarray(data_dict['labels'])
10
11 x_train, x_test, y_train, y_test = train_test_split(
12     data, labels, test_size=0.1, shuffle=True, stratify=labels
13 )
14
15
```

Run: Pelatihan dan Pengujian

Process finished with exit code 0

100.0% of samples were classified correctly!



```
14 )
15
16 model = RandomForestClassifier()
17 model.fit(x_train, y_train)
18
19 y_predict = model.predict(x_test)
20 score = accuracy_score(y_test, y_predict)
21
22 print('{}% of samples were classified correctly!'.format(score * 100))
23
24 f = open(r'C:\Users\User\Documents\Kontrol Cerdas\model.p', 'wb')
25 pickle.dump({'model': model}, f)
26 f.close()
```

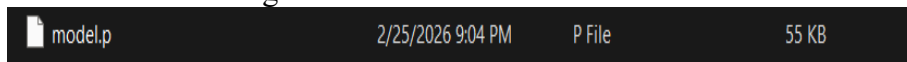
Run: Pelatihan dan Pengujian

C:\Users\User\AppData\Local\Programs\Python\Python318\python.exe "C:\Users\User\Documents\Kontrol Cerdas\idea\WEEK 2\Pelatihan dan Pengujian.py"

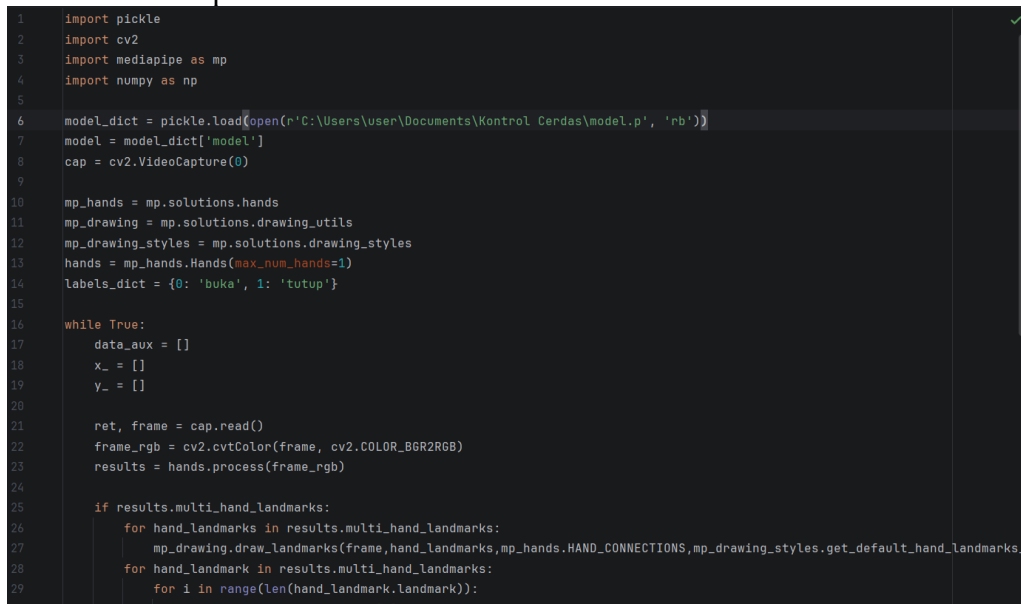
100.0% of samples were classified correctly!

Process finished with exit code 0

Dan setelah itu menghasilkan file tersebut



D. Implementasi



```
1 import pickle
2 import cv2
3 import mediapipe as mp
4 import numpy as np
5
6 model_dict = pickle.load(open(r'C:\Users\User\Documents\Kontrol Cerdas\model.p', 'rb'))
7 model = model_dict['model']
8 cap = cv2.VideoCapture(0)
9
10 mp_hands = mp.solutions.hands
11 mp_drawing = mp.solutions.drawing_utils
12 mp_drawing_styles = mp.solutions.drawing_styles
13 hands = mp_hands.Hands(max_num_hands=1)
14 labels_dict = {0: 'buka', 1: 'tutup'}
15
16 while True:
17     data_aux = []
18     x_ = []
19     y_ = []
20
21     ret, frame = cap.read()
22     frame_rgb = cv2.cvtColor(frame, cv2.COLOR_BGR2RGB)
23     results = hands.process(frame_rgb)
24
25     if results.multi_hand_landmarks:
26         for hand_landmarks in results.multi_hand_landmarks:
27             mp_drawing.draw_landmarks(frame, hand_landmarks, mp_hands.HAND_CONNECTIONS, mp_drawing_styles.get_default_hand_landmarks_
28             for hand_landmark in results.multi_hand_landmarks:
29                 for i in range(len(hand_landmark.landmark)):
30                     x = hand_landmark.landmark[i].x
```

```
25     if results.multi_hand_landmarks:
26         for hand_landmarks in results.multi_hand_landmarks:
27             mp_drawing.draw_landmarks(frame, hand_landmarks, mp_hands.HAND_CONNECTIONS, mp_drawing_styles.get_default_hand_landmarks_
28         for hand_landmark in results.multi_hand_landmarks:
29             for i in range(len(hand_landmark.landmark)):
30                 x = hand_landmark.landmark[i].x
31                 y = hand_landmark.landmark[i].y
32                 x_.append(x)
33                 y_.append(y)
34
35             for i in range(len(hand_landmark.landmark)):
36                 x = hand_landmark.landmark[i].x
37                 y = hand_landmark.landmark[i].y
38                 data_aux.append(x - min(x_))
39                 data_aux.append(y - min(y_))
40
41             prediction = model.predict([np.asarray(data_aux)])
42             predicted_character = labels_dict[int(prediction[0])]
43
44             cv2.putText(frame, predicted_character, org=(200, 100), cv2.FONT_HERSHEY_SIMPLEX, fontScale=1.3, color=(0, 0, 255), thickness=
45 cv2.imshow('winname: "frame"', frame)
46
47     if cv2.waitKey(1) & 0xFF == ord('q'):
48         break
49
50 cap.release()
51 cv2.destroyAllWindows()
```

Latihan

B. Mendeteksi orang yang mengangkat tangan posisi Landmark

```
1  import cv2
2  import mediapipe as mp
3  import numpy as np
4  import math
5
6  mp_pose = mp.solutions.pose
7  mp_drawing = mp.solutions.drawing_utils
8
9  pose = mp_pose.Pose(min_detection_confidence=0.7, min_tracking_confidence=0.7)
10
11 cap = cv2.VideoCapture(0)
12
13 def calculate_angle(a, b, c):
14     a = np.array(a)
15     b = np.array(b)
16     c = np.array(c)
17
18     radians = np.arctan2(c[1]-b[1], c[0]-b[0]) - np.arctan2(a[1]-b[1], a[0]-b[0])
19     angle = np.abs(radians*180.0/np.pi)
20
21     if angle > 180.0:
22         angle = 360-angle
23
24     return angle
25
26 while cap.isOpened():
27     success, frame = cap.read()
28     if not success:
29         break
30
31     frame = cv2.flip(frame, [flipCode: 1])
32     rgb = cv2.cvtColor(frame, cv2.COLOR_BGR2RGB)
33
34     results = pose.process(rgb)
35
36     if results.pose_landmarks:
37         landmarks = results.pose_landmarks.landmark
38
39         # Ambil koordinat kiri
40         hip = [landmarks[23].x, landmarks[23].y]
41         knee = [landmarks[25].x, landmarks[25].y]
42         ankle = [landmarks[27].x, landmarks[27].y]
43
44         angle = calculate_angle(hip, knee, ankle)
45
46         # Logika klasifikasi
47         if angle > 150:
48             status = "BERDIRI"
49             color = (0,255,0)
50         else:
51             status = "DUDUK"
52             color = (0,0,255)
53
54         mp_drawing.draw_landmarks(frame, results.pose_landmarks, mp_pose.POSE_CONNECTIONS)
55
56         cv2.putText(frame, text=f"Sudut Lutut: {int(angle)}",
57                     org=(10,50), cv2.FONT_HERSHEY_SIMPLEX,
58                     fontScale=0.7, color=(255,255,255), thickness=2)
59
60         cv2.putText(frame, text=f"Status: {status}"
```

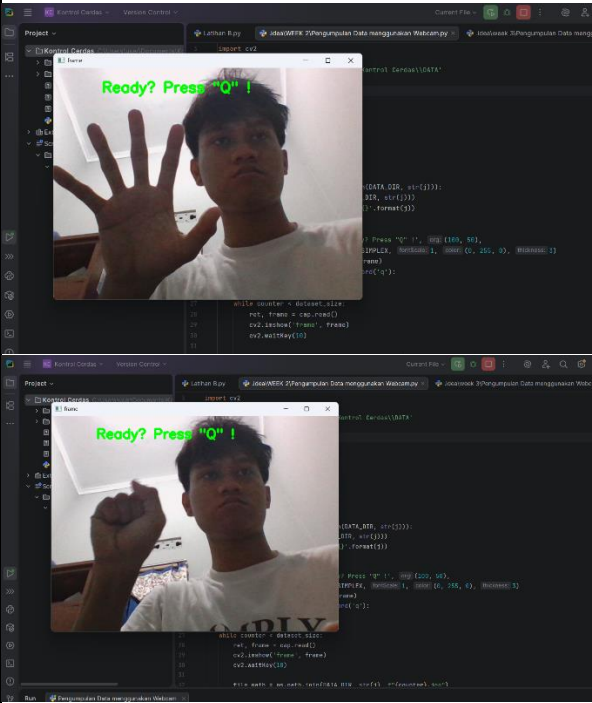
```

46         # Logika klasifikasi
47         if angle > 150:
48             status = "BERDIRI"
49             color = (0,255,0)
50         else:
51             status = "DUDUK"
52             color = (0,0,255)
53
54         mp_drawing.draw_landmarks(frame, results.pose_landmarks, mp_pose.POSE_CONNECTIONS)
55
56         cv2.putText(frame, text: f"Sudut Lutut: {int(angle)}",
57                     org: (10,50), cv2.FONT_HERSHEY_SIMPLEX,
58                     fontScale: 0.7, color: (255,255,255), thickness: 2)
59
60         cv2.putText(frame, text: f"Status: {status}",
61                     org: (10,90), cv2.FONT_HERSHEY_SIMPLEX,
62                     fontScale: 1, color, thickness: 3)
63
64         cv2.imshow( winname: "Deteksi Berdiri/Duduk", frame)
65
66         if cv2.waitKey(1) & 0xFF == 27:
67             break
68
69     cap.release()
70     cv2.destroyAllWindows()

```


Data dan Output Hasil Pengamatan :

- Hasil pengamatan berdasarkan dengan dataset baru.

No	Variabel	Hasil Pengamatan
1.	Pengumpulan data Menggunakan webcam	

2.	Ekstraksi dataset	Menghasilkan file :  data.pickle 2/25/2026 9:03 PM PICKLE File 76 KB
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3.	Pelatihan dan Pengujian	Menghasilkan file : model.p 2/25/2026 9:04 PM P File 55 KB
4	Implementasi	 This screenshot shows a person's open hand with a skeletal overlay. The word "buka" is written in red above the hand. The background is a dark IDE window with various Python files open.  This screenshot shows the same person with their hand in a closed fist gesture. The word "tutup" is written in red above the hand. The background IDE window shows the same project structure as the previous screenshot.

Daftar Pustaka :

1. Pedregosa, F., Varoquaux, G., Gramfort, A., Michel, V., Thirion, B., Grisel, O., et al. (2011). *Scikit-learn: Machine Learning in Python*. Journal of Machine Learning Research, 12, 2825–2830. (Library Scikit-learn digunakan untuk proses pelatihan dan pengujian model klasifikasi.)
2. Raschka, S., & Mirjalili, V. (2019). *Python Machine Learning: Machine Learning and Deep Learning with Python, scikit-learn, and TensorFlow*. Birmingham: Packt Publishing.
3. Géron, A. (2022). *Hands-On Machine Learning with Scikit-Learn, Keras, and TensorFlow* (3rd ed.). O'Reilly Media.
4. Documentation Team. (2024). *Scikit-learn User Guide*. Diakses dari dokumentasi resmi Scikit-learn.
5. Lugaresi, C., et al. (2019). *MediaPipe: A Framework for Building Perception Pipelines*. Framework Google ini digunakan untuk ekstraksi pose tubuh.
6. Bradski, G. (2000). *The OpenCV Library*. Dr. Dobb's Journal of Software Tools.